



Cambridge International AS & A Level

MATHEMATICS

9709/11

Paper 1 Pure Mathematics 1

May/June 2026

MARK SCHEME

Maximum Mark: 75

Published

This mark scheme contains the instructions and marking guidance that our examiners used to mark candidate responses for this component.

Before they start marking, examiners complete a standardisation process. They review a range of candidate responses and discuss the mark scheme in detail to agree which answers can and cannot be accepted. Examiners award marks where it is clear that candidate responses have met the requirements of the mark scheme. These requirements may vary between different components or different versions of the same component, depending on the exact requirements of the question and the candidate responses seen during the standardisation process.

Sometimes, our mark schemes may allow marks to be awarded to responses which contain elements that are factually incorrect or for content not covered by the syllabus. We do this in response to the demands of a specific question to support candidates and make sure that our assessments are fair. However, these allowances should not be used as a guide for teaching and learning.

We cannot discuss the mark scheme with schools, and we recommend that schools read the mark scheme together with the assessment material and the Principal Examiner Report for Teachers.

This document consists of **21** printed pages.

General marking principles

All examiners must apply these marking principles when marking candidate responses.

1	Examiners must award marks for each response in line with: - the specific content defined in the mark scheme - the specific skills defined in the mark scheme or in the level descriptions - the standard of response required as exemplified by the standardisation scripts.
2	Examiners must award whole marks (not half marks, or other fractions).
3	Examiners must award marks positively . This means that: - marks are not deducted for errors or omissions - marks are awarded for correct/valid answers as defined in the mark scheme and also for other valid answers which go beyond the scope of the syllabus and mark scheme referring to your team leader as appropriate - the quality of spelling, punctuation and grammar are judged only when the mark scheme indicates that these features are specifically assessed in the question. However, the meaning of all responses must be unambiguous.
4	Examiners must consistently apply the requirements of the mark scheme and any rules for what to do when candidates have not followed instructions correctly.
5	Examiners must use the full range of marks defined in the mark scheme, unless limited by the quality of the candidate responses seen.
6	Examiners must award marks according to the mark scheme and must not award marks with grade thresholds or grade descriptions in mind.

Mathematics-Specific Marking Principles

- 1 Unless a particular method has been specified in the question, full marks may be awarded for any correct method. However, if a calculation is required then no marks will be awarded for a scale drawing.
- 2 Unless specified in the question, non-integer answers may be given as fractions, decimals or in standard form. Ignore superfluous zeros, provided that the degree of accuracy is not affected.
- 3 Allow alternative conventions for notation if used consistently throughout the paper, e.g. commas being used as decimal points.
- 4 Unless otherwise indicated, marks once gained cannot subsequently be lost, e.g. wrong working following a correct form of answer is ignored (isw).
- 5 Where a candidate has misread a number or sign in the question and used that value consistently throughout, provided that number does not alter the difficulty or the method required, award all marks earned and deduct just 1 A or B mark for the misread.
- 6 Recovery within working is allowed, e.g. a notation error in the working where the following line of working makes the candidate's intent clear.

Annotations guidance for centres

Examiners use a system of annotations as a shorthand for communicating their marking decisions to one another. Examiners are trained during the standardisation process on how and when to use annotations. The purpose of annotations is to inform the standardisation and monitoring processes and guide the supervising examiners when they are checking the work of examiners within their team. The meaning of annotations and how they are used is specific to each component and is understood by all examiners who mark the component.

We publish annotations in our mark schemes to help centres understand the annotations they may see on copies of scripts. Note that there may not be a direct correlation between the number of annotations on a script and the mark awarded. Similarly, the use of an annotation may not be an indication of the quality of the response.

The annotations listed below were available to examiners marking this component in this series.

Annotations

Annotation	Meaning
	More information required
	Accuracy mark awarded zero
	Accuracy mark awarded one
	Independent accuracy mark awarded zero
	Independent accuracy mark awarded one
	Independent accuracy mark awarded two
	Benefit of the doubt
	Blank Page
	Incorrect
Dep	Used to indicate DM0 or DM1

Annotation	Meaning
DM1	Dependent on the previous M1 mark(s)
FT	Follow through
	Indicate working that is right or wrong
Highlighter	Highlight a key point in the working
ISW	Ignore subsequent work
J	Judgement
JU	Judgement
M0	Method mark awarded zero
M1	Method mark awarded one
M2	Method mark awarded two
MR	Misread
O	Omission or Other solution
Off-page comment	Allows comments to be entered at the bottom of the RM marking window and then displayed when the associated question item is navigated to.
On-page comment	Allows comments to be entered in speech bubbles on the candidate response.
PE	Judgment made by the PE
Pre	Premature approximation
SC	Special case
SEEN	Indicates that work/page has been seen

Annotation	Meaning
	Error in number of significant figures
	Correct
	Transcription error
	Correct answer from incorrect working

PUBLISHED**Mark Scheme Notes**

The following notes are intended to aid interpretation of mark schemes in general, but individual mark schemes may include marks awarded for specific reasons outside the scope of these notes.

Types of mark

- M** Method mark, awarded for a valid method applied to the problem. Method marks are not lost for numerical errors, algebraic slips or errors in units. However, it is not usually sufficient for a candidate just to indicate an intention of using some method or just to quote a formula; the formula or idea must be applied to the specific problem in hand, e.g. by substituting the relevant quantities into the formula. Correct application of a formula without the formula being quoted obviously earns the M mark and in some cases an M mark can be implied from a correct answer.
- A** Accuracy mark, awarded for a correct answer or intermediate step correctly obtained. Accuracy marks cannot be given unless the associated method mark is earned (or implied).
- B** Mark for a correct result or statement independent of method marks.
- DM or DB** When a part of a question has two or more ‘method’ steps, the M marks are generally independent unless the scheme specifically says otherwise; and similarly, when there are several B marks allocated. The notation DM or DB is used to indicate that a particular M or B mark is dependent on an earlier M or B (asterisked) mark in the scheme. When two or more steps are run together by the candidate, the earlier marks are implied and full credit is given.
- FT** Implies that the A or B mark indicated is allowed for work correctly following on from previously incorrect results. Otherwise, A or B marks are given for correct work only.
- A or B marks are given for correct work only (not for results obtained from incorrect working) unless follow through is allowed (see abbreviation FT above).
 - For a numerical answer, allow the A or B mark if the answer is correct to 3 significant figures or would be correct to 3 significant figures if rounded (1 decimal place for angles in degrees).
 - The total number of marks available for each question is shown at the bottom of the Marks column.
 - Wrong or missing units in an answer should not result in loss of marks unless the guidance indicates otherwise.
 - Square brackets [] around text or numbers show extra information not needed for the mark to be awarded.

Abbreviations

AEF/OE Any Equivalent Form (of answer is equally acceptable) / Or Equivalent

AG Answer Given on the question paper (so extra checking is needed to ensure that the detailed working leading to the result is valid)

CAO Correct Answer Only (emphasising that no ‘follow through’ from a previous error is allowed)

CWO Correct Working Only

ISW Ignore Subsequent Working

SOI Seen Or Implied

SC Special Case (detailing the mark to be given for a specific wrong solution, or a case where some standard marking practice is to be varied in the light of a particular circumstance)

WWW Without Wrong Working

AWRT Answer Which Rounds To

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Question	Answer	Marks	Guidance
1	EITHER Solution 1 $r^2 = \frac{12.5}{50} \quad [\Rightarrow r = \frac{1}{2}]$	(M1)	Correct procedure to find r^2 or r .
	$a = 200$ and $r = \frac{1}{2}$	A1	Ignore $r = -\frac{1}{2}$.
	$S_{\infty} = \frac{200}{1 - \frac{1}{2}}$	M1	Use of formula for sum to infinity for <i>their</i> a and r provided that $ r < 1$.
	$S_{\infty} = 400$	A1)	CAO
	OR Solution 2 Alternative method for M1 A1 $a = \frac{2500}{12.5} \quad [\Rightarrow a = 200]$	(M1)	Correct procedure to find a .
	$r = \frac{1}{2}$ and $a = 200$	A1)	
	Available marks	4	

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Question	Answer	Marks	Guidance
2(a)(i)	$(1+2x)^5 = 1+10x+40x^2$	B1	B1 for any two.
		B1	B1 for all three. ISW for extra terms.
		2	
2(a)(ii)	$(3+x^2)^6 = 729+1458x^2$	B1	ISW extra terms, e.g. $1215x^4$.
		1	
2(b)	Coefficient of x^2 is $1 \times 1458 + 40 \times 729$	M1	For using <i>their</i> values to find coefficient from $(ax^2 + bx + c)(d + ex^2) \Rightarrow ad + ce$.
	30618	A1FT	Condone $30618x^2$.
		2	

Question	Answer	Marks	Guidance
3	$y = \left(\frac{x-5}{4}\right)^3 - 2$	B1	B1 for $x-5$, or subtracting 2 or dividing by 4^3 .
		B1	Second B1 for another correct procedure.
		B1	Final B1 for fully correct OE, e.g. $y = \frac{1}{64}(x-5)^3 - 2$.
		3	

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Question	Answer	Marks	Guidance
4	EITHER Solution 1 $8\sin^4\theta - 6(1 - \sin^2\theta) + 1 = 0 \quad [\Rightarrow 8\sin^4\theta + 6\sin^2\theta - 5 = 0]$	(M1)*	Use of $\sin^2\theta + \cos^2\theta = 1$ to eliminate $\cos^2\theta$.
	$(4\sin^2\theta + 5)(2\sin^2\theta - 1) = 0 \quad [\Rightarrow \sin^2\theta = \frac{1}{2} \text{ or } \sin^2\theta = -\frac{5}{4}]$	DM1	Solve <i>their</i> three-term quadratic in $\sin^2\theta$ using suitable method, e.g. factorisation, quadratic formula, completing the square.
	$\sin\theta = \pm \frac{1}{\sqrt{2}}$	A1	SC B1 if M1 M0, allow AWRT ± 0.707 .
	$\theta = \frac{1}{4}\pi, \frac{3}{4}\pi, \frac{5}{4}\pi, \frac{7}{4}\pi$ or 0.785, 2.36, 3.93, 5.50	A1	A1 for any correct pair.
		A1)	A1 for all four and no other solutions in range. SC B1 for 45°, 135°, 225°, 315°.
	OR Solution 2 Alternative Method for Question 4 $8(1 - \cos^2\theta)^2 - 6\cos^2\theta + 1 = 0 \quad [\Rightarrow 8\cos^4\theta - 22\cos^2\theta + 9 = 0]$	(M1)*	Use of $\sin^2\theta + \cos^2\theta = 1$ to eliminate $\sin^2\theta$.
	$(4\cos^2\theta - 9)(2\cos^2\theta - 1) = 0 \quad [\Rightarrow \cos^2\theta = \frac{9}{4} \text{ or } \cos^2\theta = \frac{1}{2}]$	DM1	Solve three-term quadratic in $\cos^2\theta$ using suitable method, e.g. factorisation, quadratic formula, completing the square.
	$\cos\theta = \pm \frac{1}{\sqrt{2}}$	A1	SC B1 if M1 M0, allow AWRT ± 0.707 .
	$\theta = \frac{1}{4}\pi, \frac{3}{4}\pi, \frac{5}{4}\pi, \frac{7}{4}\pi$	A1	A1 for any correct pair.
		A1)	A1 for all four and no other solutions in range. SC B1 for 45°, 135°, 225°, 315°.
	Available marks	5	

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Question	Answer	Marks	Guidance
5	$\left[\frac{dy}{dx}\right] = -0.5 + \frac{1}{2}(x^3 - 3x - 2)^{-\frac{1}{2}}(3x^2 - 3)$	B1B1	B1 for $\frac{1}{2}(x^3 - 3x - 2)^{-\frac{1}{2}}$, B1 for the rest.
	At $x = 6$, $\left[\frac{dy}{dx}\right] = -0.5 + \frac{1}{2} \times 196^{-\frac{1}{2}} \times 105 \left[= \frac{13}{4} \right]$	M1	For attempt at gradient by substituting $x = 6$ into <i>their</i> $\frac{dy}{dx}$.
	[Gradient of normal] = $-\frac{4}{13}$	M1	For use of product of gradients = -1 using <i>their</i> $\frac{dy}{dx}$.
	Equation is $4x + 13y - 167 = 0$	A1	OE in required form.
		5	

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Question	Answer	Marks	Guidance
6	EITHER Solution 1 $p^2 + (p+6) = 2 \times 4p$ or $4p - (p+6) = p^2 - 4p$ $[\Rightarrow p^2 - 7p + 6 = 0]$	(M1)*	OE Correct method for forming quadratic equation. Allow one sign error.
	$p = 1$ and $p = 6$	A1	
	For $p = 1$, $d = -3$, or $p = 6$, $d = 12$	DM1	For finding d for one progression.
	$a = 7 - (-3) \times 9 [= 34]$ $a = 12 - 12 \times 9 [= -96]$	DM1	For finding a for one progression.
	$u_{20} = 34 - 3 \times 19 = -23$ $u_{20} = -96 + 19 \times 12 = 132$	A1	For both.
	Difference = 155	A1)	Allow -155.
	OR Solution 2 Alternative method for 2nd and 3rd M1 Lists at least three consecutive terms with position clear	(DM1	E.g., 7, 4, 1 are 10th, 11th, 12th terms or 34, 31, 28 are 1st, 2nd, 3rd terms.
	Lists at least three consecutive terms with position clear	DM1)	E.g. 12, 24, 36 are 10th, 11th, 12th terms, or -96, -84, -72 are 1st, 2nd, 3rd terms.
	Available marks	6	

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Question	Answer	Marks	Guidance
7(a)	$(r\sqrt{7})^2 = r^2 + (2r)^2 - 2 \times 2r \times r \times \cos \alpha \left[\Rightarrow \alpha = \cos^{-1} \frac{4+1-7}{2 \times 2 \times 1} \right]$	M1	Correct use of cosine rule.
	$= \frac{2}{3}\pi$ or 120°	A1	
	Arc length $= 2r \times \frac{2}{3}\pi$ or $\frac{120}{360} \times 2\pi \times 2r$	M1	Use of $r\theta$ with <i>their</i> angle. Using <i>their</i> numeric angle in radians.
	Total perimeter $= \left(\frac{4}{3}\pi + 1 + \sqrt{7}\right)r$	A1	Or exact equivalent. Allow unsimplified.
		4	
7(b)	Sector area $= \frac{1}{2} \times (2r)^2 \times \frac{2}{3}\pi$ or $\frac{120}{360} \times \pi \times (2r)^2$	M1*	Using <i>their</i> numeric angle.
	Triangle area $= \frac{1}{2} \times r \times 2r \times \sin \frac{2}{3}\pi$ or $= \frac{1}{2} \times r \times 2r \times \sin 120$	M1*	Using <i>their</i> numeric angle.
	Area of shaded section $= \frac{1}{2} \times (2r)^2 \times \frac{2}{3}\pi - \frac{1}{2} \times r \times 2r \times \sin \frac{2}{3}\pi$	DM1	For subtraction of <i>their</i> areas.
	$= \left(\frac{4}{3}\pi - \frac{\sqrt{3}}{2}\right)r^2$	A1	Or exact equivalent. Allow unsimplified with $\sin \frac{2}{3}\pi$ evaluated.
		4	

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Question	Answer	Marks	Guidance
8	Meet when $\frac{5}{2x-1} = 7-2x \Rightarrow -4x^2 + 16x - 12 = [0] [\Rightarrow x^2 - 4x + 3 = 0]$	M1	For expressing as a three-term quadratic. All terms need not be on the same side.
	$x = 1$ and $x = 3$	A1	
	Volume = $[\pi] \int \left((7-2x)^2 - \left(\frac{5}{2x-1} \right)^2 \right) dx$	M1*	Subtracting two volumes. Condone subtraction the wrong way round.
	$[\pi] \left\{ \left\{ 49x - 14x^2 + \frac{4}{3}x^3 \right\} [-] \left\{ -\frac{1}{2} \right\} \left\{ \frac{25}{2x-1} \right\} \right\}$	B1	Or $\left\{ -\frac{1}{6}(7-2x)^3 \right\}$. Allow unsimplified.
		B1	Allow unsimplified.
		B1	Allow unsimplified.
	$[\pi] \left(57 + \frac{5}{2} \right) - [\pi] \left(\frac{109}{3} + \frac{25}{2} \right)$ Minimum acceptable	DM1	Use of <i>their</i> limits in an attempt at integration. Condone one sign error or missing brackets. Note $(49 \times 3 - 14 \times 3^2 + \frac{4}{3} \times 3^3 + \frac{1}{2} \left(\frac{25}{2 \times 3 - 1} \right)) - (49 \times 1 - 14 \times 1^2 + \frac{4}{3} \times 1^3 + \frac{1}{2} \left(\frac{25}{2 \times 1 - 1} \right))$ $= (147 - 126 + 36 + \frac{5}{2}) - (\frac{109}{3} + \frac{25}{2})$ $= (57 + \frac{5}{2}) - (\frac{109}{3} + \frac{25}{2})$ $= \frac{119}{2} - \frac{293}{6}$ Or $\frac{62}{3} - 10$

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Question	Answer	Marks	Guidance
8	Volume = $\frac{32}{3}\pi$	B1	Correct limits must be clear and integration correct, WWW.
		8	

Question	Answer	Marks	Guidance
9(a)	$y^3 = x \pm 8a$ or $x^3 = y \pm 8a$	M1	
	$[g^{-1}(x)] = \sqrt[3]{x+8a}$	A1	Do not allow “ $x = \dots$ ”.
		2	
9(b)(i)	$f(-1) = -4$	B1	
	$g(-4) = (-4)^3 - 8a \Rightarrow (-4)^3 - 8a + 20a^2 [= 0]$	M1	For $g(\text{their 'f(-1)'})$ and forming quadratic equation with all terms on one side.
	$4(5a+8)(a-2)$	M1	Solving <i>their</i> quadratic using suitable method, e.g. factorisation, quadratic formula, completing the square. M0 if roots are not real.
	$a = 2$	A1	SC B1 if M1M0. Ignore answer of $a = -\frac{8}{5}$.
		4	

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Question	Answer	Marks	Guidance
9(b)(ii)	For either $gf(x) = (x-3)^3 - 16$ or $fg(x) = (x^3 - 16) - 3$	B1FT	In correct part of equation with $a > 0$. May be implied by correct equation. Allow $gf(x) = (x-3)^3 - 8a$ or $fg(x) = (x^3 - 8a) - 3$.
	$(x-3)^3 - 16 = (x^3 - 16) - 3 + k$	M1*	For forming equation with a substituted with a real or $8a$ eliminated. gf and fg must be the correct way round
	$x^3 - 9x^2 + 27x - 27 - 16 = x^3 - 19 + k \Rightarrow -9x^2 + 27x - 24 - k [= 0]$	DM1	For multiplying out and simplifying. All terms to one side. Allow one sign error only.
	$27^2 - 4 \times 9 \times (24 + k) [< 0]$	M1	For use of $b^2 - 4ac$ correct for <i>their</i> a, b, c from a three-term quadratic and including k .
	$k > -3.75$ or $k > -\frac{15}{4}$ or $k > -\frac{135}{36}$	A1	OE
		5	

Question	Answer	Marks	Guidance
10(a)	At $x = 2$, $\frac{d^2y}{dx^2} = 12 - 6$ [> 0] $\Rightarrow$ minimum	B1	
		1	

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Question	Answer	Marks	Guidance
10(b)	$\frac{dy}{dx} = x^3 - 6x + c$	B1	$\frac{dy}{dx}$ can be implied by later work. Allow $f'(x)$ or y' .
	$0 = 8 - 12 + c \Rightarrow c = 4 \Rightarrow \frac{dy}{dx} = x^3 - 6x + 4$	M1	For substituting $\frac{dy}{dx} = 0$ at $x = 2$.
	$y = \frac{1}{4}x^4 - 3x^2 + 4x + k$	A1	Can be implied by later work.
	$3 = 4 - 12 + 8 + k \Rightarrow k = 3$	M1	For substituting (2, 3) into <i>their</i> “ $y = \dots$ ”
	$y = \frac{1}{4}x^4 - 3x^2 + 4x + 3$	A1	Must be an equation unless $y =$ seen earlier in the question. Allow “ $f(x) = \dots$ ”.
		5	
10(c)(i)	$x^3 - 6x + 4 = (x - 2)(x^2 + 2x - 2) [= 0]$	M1*	For using <i>their</i> k in the given relationship or factorising otherwise.
		DM1	Solving quadratic using suitable method, e.g. quadratic formula.
	$x = -1 \pm \sqrt{3}$	A1	SC B1 if M1 M0.
		3	
10(c)(ii)	At $x = -1 + \sqrt{3} \quad \left[\frac{d^2y}{dx^2} \right] = 12 - 6\sqrt{3} - 6$ or $-4.39\dots [< 0] \Rightarrow$ maximum	B1	Must see $\frac{d^2y}{dx^2}$ evaluated.
	At $x = -1 - \sqrt{3} \quad \left[\frac{d^2y}{dx^2} \right] = 6\sqrt{3} + 12 - 6$ or $16.4\dots [> 0] \Rightarrow$ minimum	B1	Must see $\frac{d^2y}{dx^2}$ evaluated.
		2	

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Question	Answer	Marks	Guidance
11	EITHER Solution 1 Gradients of normal lines are $\frac{16-10}{6-8} [= -3]$ and $\frac{8-10}{14-8} [= -\frac{1}{3}]$	(M1)*	For either normal gradient.
	Gradients of tangent lines are $\frac{1}{3}$ and 3	DM1	For either tangent gradient. Must be clear that they are finding tangent gradient.
	Equation of tangent at (6, 16) is $y - 16 = \frac{1}{3}(x - 6) [y = \frac{1}{3}x + 14]$	A1	
	Equation of tangent at (14, 8) is $y - 8 = 3(x - 14) [y = 3x - 34]$	A1	
	Meet when $3x - 34 = \frac{1}{3}x + 14 \Rightarrow 8x = 144 [\Rightarrow x = 18]$	M1	Equating <i>their</i> two tangents.
	$x = 18$	A1	
	$y = 3x - 34$ meets x -axis at $\frac{34}{3}$ or $y = \frac{1}{3}x + 14$ meets y -axis at 14	B1	For either. Coordinates of P or R must be clear or indication of where the lines cross the x - or y -axes.
	Finds y -coordinate of Q and uses Area = rectangle – triangles $= 18 \times 20 - \frac{1}{2} \times 18 \times (20 - 14) - \frac{1}{2} \times \left(18 - \frac{34}{3}\right) \times 20$	M1	For correct attempt at area, e.g. trapezium – triangle, integration.
	$= 360 - 54 - \frac{200}{3} = \frac{718}{3}$ or $239\frac{1}{3}$	A1)	

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Question	Answer	Marks	Guidance
11	OR Solution 2 Alternative method for M1 M1 $\frac{dy}{dx} = \frac{16-2x}{2y-20}$	(M1*	
	$\frac{16-2 \times 6}{2 \times 16-20} \left[= \frac{1}{3} \right] \text{ or } \frac{16-2 \times 14}{2 \times 8-20} [= 3]$	DM1)	
	OR Solution 3 Alternative method 1 for final M1 A1 Finds y-coordinate of Q and uses Area = trapezium – triangle $= \left(\frac{14+20}{2} \right) \times 20 - \frac{\left(18 - \frac{34}{3} \right) \times 20}{2}$	(M1	
	$= 306 - \frac{200}{3} = \frac{718}{3} \text{ or } 239\frac{1}{3}$	A1)	
	OR Solution 2 Alternative method 2 for final M1 A1 $\int_0^{18} \left(\frac{1}{3}x + 14 \right) dx - \int_{\frac{34}{3}}^{18} (3x - 34) dx \Rightarrow \frac{1}{6}x^2 + 14x - \left(\frac{3}{2}x^2 - 34x \right)$	(M1	
	$306 - \frac{200}{3} = \frac{718}{3} \text{ or } 239\frac{1}{3}$	A1)	

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Question	Answer	Marks	Guidance
11	OR Solution 2 Alternative method 1 for final M1 M1 A1 A1 $(x - 8)(x_1 - 8) + (y - 10)(y_1 - 10) = 40$	(M1)	May be implied by later work.
	Substitutes (6, 16) or (14, 8)	DM1	
	$-2(x - 8) + 6(y - 10) = 40 \Rightarrow y = \frac{1}{3}x + 14$	A1	
	$6(x - 8) - 2(y - 10) = 40 \Rightarrow y = 3x - 34$	A1)	
	Available marks	9	